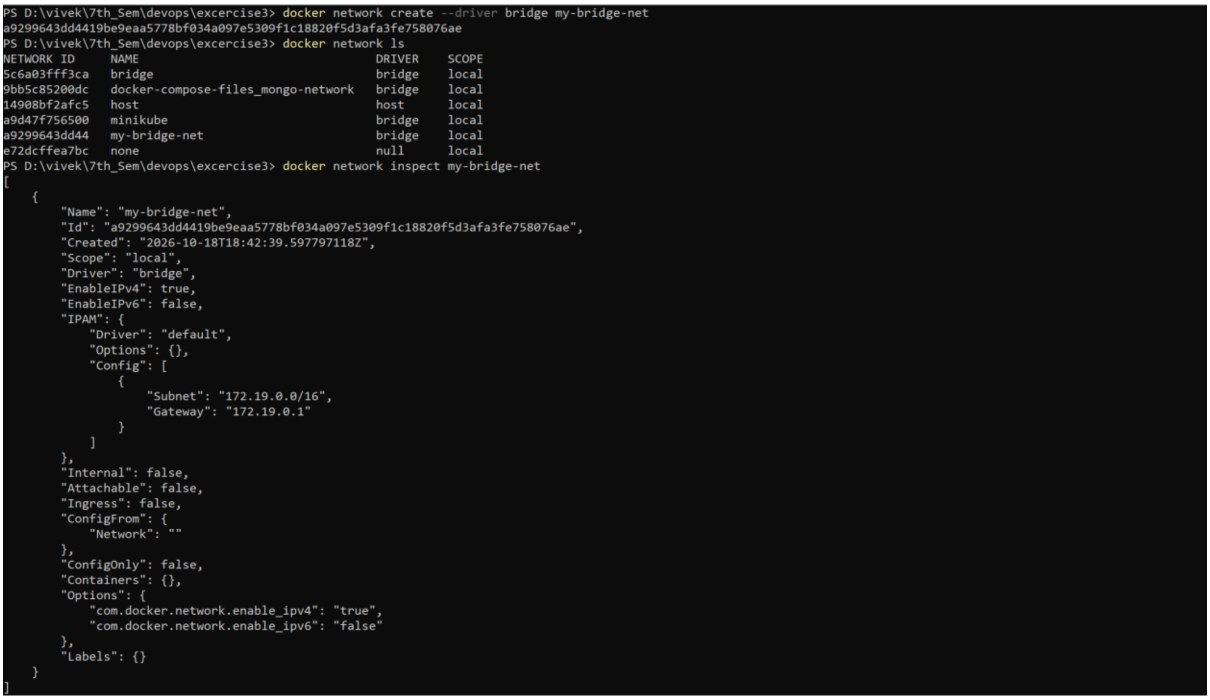
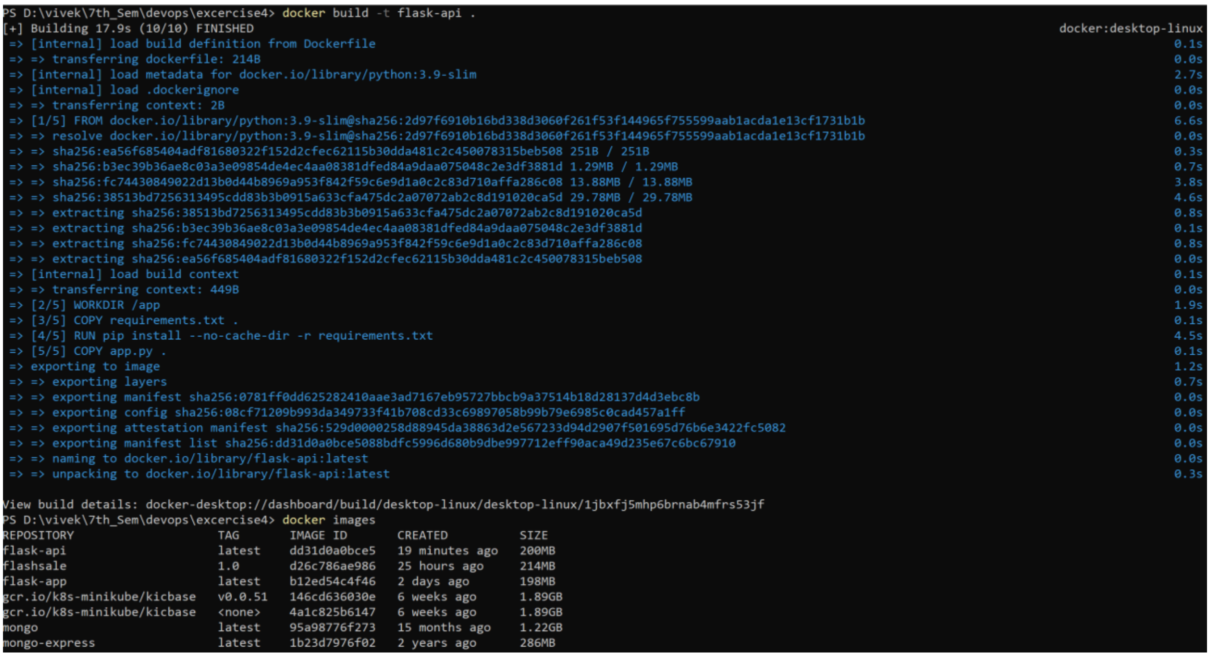
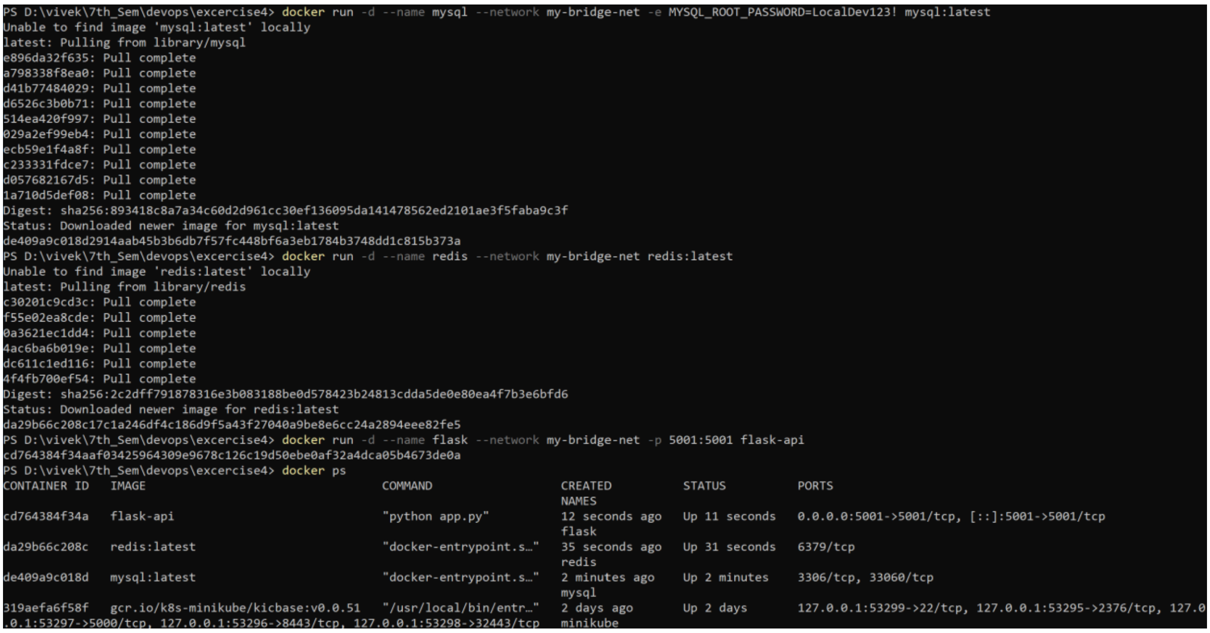
Devops – Exercise 4



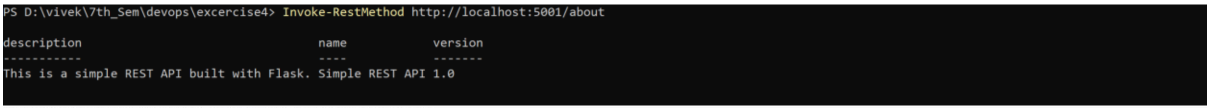
```
PS D:\vivek\7th_Sem\devops\excercise3> docker network create --driver bridge my-bridge-net
a9299643dd4419be9eaa5778bf034a097e5309f1c18820f5d3afa3fe758076ae
PS D:\vivek\7th_Sem\devops\excercise3> docker network ls
NETWORK ID     NAME                                 DRIVER    SCOPE
5c6a03fff3ca   bridge                               bridge    local
9bb5c85200dc   docker-compose-files_mongo-network   bridge    local
14908bf2afc5   host                                 host      local
a9d47f756500   minikube                             bridge    local
a9299643dd44   my-bridge-net                        bridge    local
e72dcffea7bc   none                                 null      local
PS D:\vivek\7th_Sem\devops\excercise3> docker network inspect my-bridge-net
[
    {
        "Name": "my-bridge-net",
        "Id": "a9299643dd4419be9eaa5778bf034a097e5309f1c18820f5d3afa3fe758076ae",
        "Created": "2026-10-18T18:42:39.597797118Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {}
    }
]
```



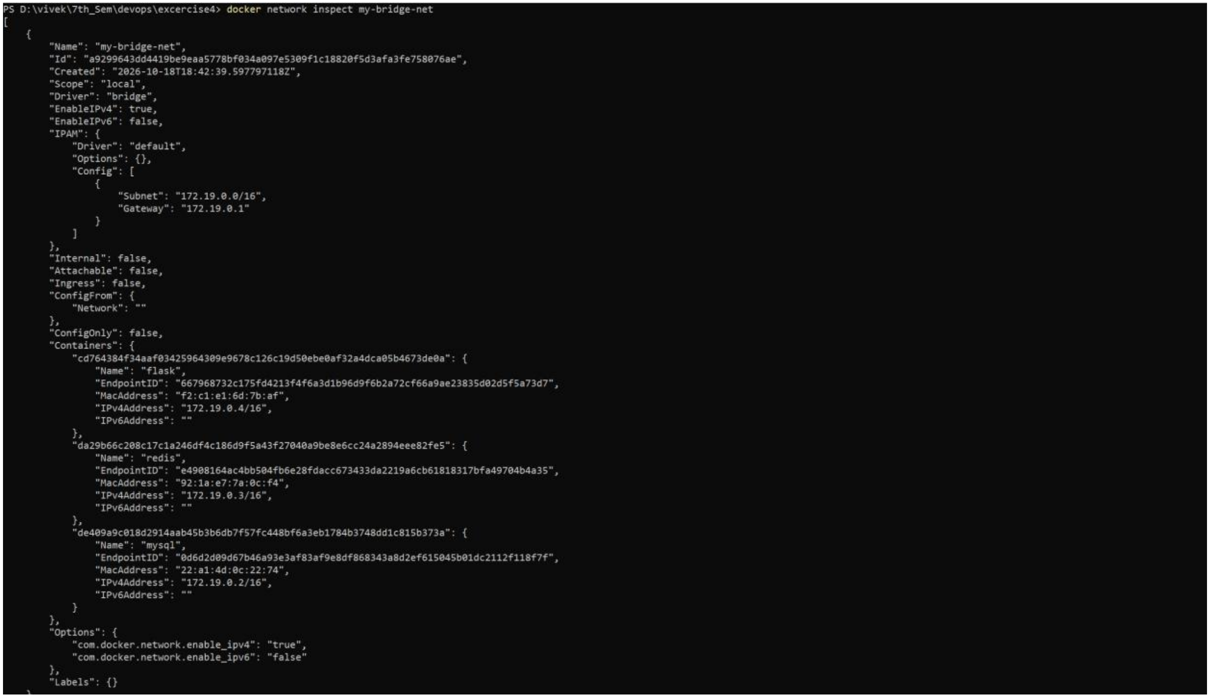
```
PS D:\vivek\7th_Sem\devops\excercise4> docker build -t flask-api .
[+] Building 17.9s (10/10) FINISHED                                                                                                                 docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                                                0.1s
 => => transferring dockerfile: 214B                                                                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                                                                  2.7s
 => [internal] load .dockerignore                                                                                                                                   0.0s
 => => transferring context: 2B                                                                                                                                     0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b                                            6.6s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b                                            0.0s
 => => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B                                                                          0.3s
 => => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB                                                                      0.7s
 => => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB                                                                    3.8s
 => => sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB                                                                    4.6s
 => => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d                                                                           0.8s
 => => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d                                                                           0.1s
 => => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08                                                                           0.8s
 => => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508                                                                           0.0s
 => [internal] load build context                                                                                                                                   0.1s
 => => transferring context: 449B                                                                                                                                   0.0s
 => [2/5] WORKDIR /app                                                                                                                                              1.9s
 => [3/5] COPY requirements.txt .                                                                                                                                   0.1s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                                                                        4.5s
 => [5/5] COPY app.py .                                                                                                                                             0.1s
 => exporting to image                                                                                                                                              1.2s
 => => exporting layers                                                                                                                                             0.7s
 => => exporting manifest sha256:0781ff0dd625282410aae3ad7167eb95727bbcb9a37514b18d28137d4d3ebc8b                                                                   0.0s
 => => exporting config sha256:08cf71209b993da349733f41b708cd33c69897058b99b79e6985c0cad457a1ff                                                                     0.0s
 => => exporting attestation manifest sha256:529d0000258d88945da38863d2e567233d94d2907f501695d76b6e3422fc5082                                                       0.0s
 => => exporting manifest list sha256:dd31d0a0bce5088bdfc5996d680b9dbe997712eff90aca49d235e67c6bc67910                                                              0.0s
 => => naming to docker.io/library/flask-api:latest                                                                                                                 0.0s
 => => unpacking to docker.io/library/flask-api:latest                                                                                                              0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/1jbxfj5mhp6brnab4mfrs53jf
PS D:\vivek\7th_Sem\devops\excercise4> docker images
REPOSITORY                    TAG       IMAGE ID       CREATED          SIZE
flask-api                     latest    dd31d0a0bce5   19 minutes ago   200MB
flashsale                     1.0       d26c786ae986   25 hours ago     214MB
flask-app                     latest    b12ed54c4f46   2 days ago       198MB
gcr.io/k8s-minikube/kicbase   v0.0.51   146cd636030e   6 weeks ago      1.89GB
gcr.io/k8s-minikube/kicbase   <none>    4a1c825b6147   6 weeks ago      1.89GB
mongo                         latest    95a98776f273   15 months ago    1.22GB
mongo-express                 latest    1b23d7976f02   2 years ago      286MB
```



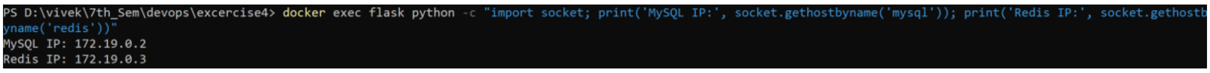
```
PS D:\vivek\7th_Sem\devops\excercise4> docker run -d --name mysql --network my-bridge-net -e MYSQL_ROOT_PASSWORD=LocalDev123! mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
e896da32f635: Pull complete
a798338f8ea0: Pull complete
d41b77484029: Pull complete
d6526c3b0b71: Pull complete
514ea420f997: Pull complete
029a2ef99eb4: Pull complete
ecb59e1f4a8f: Pull complete
c233331fdce7: Pull complete
d057682167d5: Pull complete
1a710d5def08: Pull complete
Digest: sha256:893418c8a7a34c60d2d961cc30ef136095da141478562ed2101ae3f5faba9c3f
Status: Downloaded newer image for mysql:latest
de409a9c018d2914aab45b3b6db7f57fc448bf6a3eb1784b3748dd1c815b373a
PS D:\vivek\7th_Sem\devops\excercise4> docker run -d --name redis --network my-bridge-net redis:latest
Unable to find image 'redis:latest' locally
latest: Pulling from library/redis
c30201c9cd3c: Pull complete
f55e02ea8cde: Pull complete
0a3621ec1dd4: Pull complete
4ac6ba6b019e: Pull complete
dc611c1ed116: Pull complete
4f4fb700ef54: Pull complete
Digest: sha256:2c2dff791878316e3b083188be0d578423b24813cdda5de0e80ea4f7b3e6bfd6
Status: Downloaded newer image for redis:latest
da29b66c208c17c1a246df4c186d9f5a43f27040a9be8e6cc24a2894eee82fe5
PS D:\vivek\7th_Sem\devops\excercise4> docker run -d --name flask --network my-bridge-net -p 5001:5001 flask-api
cd764384f34aaf03425964309e9678c126c19d50ebe0af32a4dca05b4673de0a
PS D:\vivek\7th_Sem\devops\excercise4> docker ps
CONTAINER ID   IMAGE                             COMMAND                  CREATED          STATUS          PORTS
                                                                          NAMES
cd764384f34a   flask-api                         "python app.py"          12 seconds ago   Up 11 seconds   0.0.0.0:5001->5001/tcp, [::]:5001->5001/tcp
                                                                          flask
da29b66c208c   redis:latest                      "docker-entrypoint.s…"   35 seconds ago   Up 31 seconds   6379/tcp
                                                                          redis
de409a9c018d   mysql:latest                      "docker-entrypoint.s…"   2 minutes ago    Up 2 minutes    3306/tcp, 33060/tcp
                                                                          mysql
319aefa6f58f   gcr.io/k8s-minikube/kicbase:v0.0.51   "/usr/local/bin/entr…"   2 days ago       Up 2 days       127.0.0.1:53299->22/tcp, 127.0.0.1:53295->2376/tcp, 127.0
.0.1:53297->5000/tcp, 127.0.0.1:53296->8443/tcp, 127.0.0.1:53298->32443/tcp   minikube
```



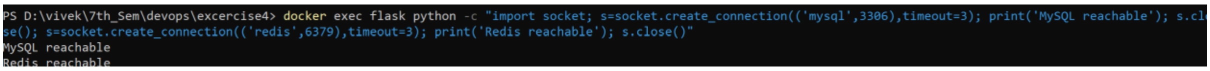
```
PS D:\vivek\7th_Sem\devops\excercise4> Invoke-RestMethod http://localhost:5001/about

description                            name            version
-----------                            ----            -------
This is a simple REST API built with Flask. Simple REST API 1.0
```



```
PS D:\vivek\7th_Sem\devops\excercise4> docker network inspect my-bridge-net
[
    {
        "Name": "my-bridge-net",
        "Id": "a9299643dd4419be9eaa5778bf034a097e5309f1c18820f5d3afa3fe758076ae",
        "Created": "2026-10-18T18:42:39.597797118Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {
            "cd764384f34aaf03425964309e9678c126c19d50ebe0af32a4dca05b4673de0a": {
                "Name": "flask",
                "EndpointID": "667968732c175fd4213f4f6a3d1b96d9f6b2a72cf66a9ae23835d02d5f5a73d7",
                "MacAddress": "f2:c1:e1:6d:7b:af",
                "IPv4Address": "172.19.0.4/16",
                "IPv6Address": ""
            },
            "da29b66c208c17c1a246df4c186d9f5a43f27040a9be8e6cc24a2894eee82fe5": {
                "Name": "redis",
                "EndpointID": "e4908164ac4bb504fb6e28fdacc673433da2219a6cb61818317bfa49704b4a35",
                "MacAddress": "92:1a:e7:7a:0c:f4",
                "IPv4Address": "172.19.0.3/16",
                "IPv6Address": ""
            },
            "de409a9c018d2914aab45b3b6db7f57fc448bf6a3eb1784b3748dd1c815b373a": {
                "Name": "mysql",
                "EndpointID": "0d6d2d09d67b46a93e3af83af9e8df868343a8d2ef615045b01dc2112f118f7f",
                "MacAddress": "22:a1:4d:0c:22:74",
                "IPv4Address": "172.19.0.2/16",
                "IPv6Address": ""
            }
        },
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {}
```



```
PS D:\vivek\7th_Sem\devops\excercise4> docker exec flask python -c "import socket; print('MySQL IP:', socket.gethostbyname('mysql')); print('Redis IP:', socket.gethostb
yname('redis'))"
MySQL IP: 172.19.0.2
Redis IP: 172.19.0.3
```



```
PS D:\vivek\7th_Sem\devops\excercise4> docker exec flask python -c "import socket; s=socket.create_connection(('mysql',3306),timeout=3); print('MySQL reachable'); s.cl
se(); s=socket.create_connection(('redis',6379),timeout=3); print('Redis reachable'); s.close()"
MySQL reachable
Redis reachable
```